# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Data Science

## CO1-ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

| | | | |
|---|---|---|---|
| **Course Code:** | DSA03 | **Course Title:** | Natural Language Processing |
| **CO Assessed:** | CO1 – Imitation (L1) | **Assessment:** | Assessment Tool 1 – Application Based Questions |
| **Weightage:** | 40% of CIA | | |
| **CO1 Statement:** | Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3) | | |
| **Student Name:** | S.Bhanu | **Reg. No.:** | 192425195 |
| **Section / Batch:** | 2024 | **Date:** | 24-10-2026 |

**Code of Conduct**

*I* S.Bhanu(192425195) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: S.Bhanu

**Instructions**

- Develop a modular and efficient Python program that satisfies all the functional requirements specified in the problem statement.
- Demonstrate the correctness of the implementation using suitable test cases and justify the output obtained.
- Follow good programming practices, including meaningful variable names, proper code indentation, comments, and error handling wherever applicable.
- Duration: 30 minutes.

| Section / Topic | Focus | Q Nos |
|---|---|---|
| **Information Extraction** | Regex-Based Information Extraction | 1 |
| **Pattern Matching** | Regex Pattern Matching | 2 |
| **Regular Expression Patterns** | Regex-Based Input Validation | 3 |

## Annexure A – Application Based Questions

### Section 1: Regular Expressions – Resume Information Extraction

1. A multinational recruitment company receives thousands of resumes every day in text format. The HR department requires an automated system to extract candidate information for shortlisting.

**Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:**

Extract the candidate's name.
Identify email addresses.
Extract mobile numbers.
Detect technical skills (Python, Java, SQL, Machine Learning, NLP).
Extract years of experience.
Generate a structured summary of the candidate's profile.
Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

### Section 2: Regular Expressions – Product Search System

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

**Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:**

Search products using an exact keyword.
Perform prefix-based searches.
Perform suffix-based searches.
Search using partial keywords.
Perform case-insensitive searches.
Display all matching product names.
Generate a report showing the total number of matching products for each search.

### Section 3: Regular Expressions – University Registration System

A university is developing an automated online registration portal to verify student information before course enrollment.

**Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:**

Validate the student register number.
Validate the institutional email address.
Validate the course code.
Validate the semester information.
Validate the student's mobile number.
Display appropriate validation messages for each field.
Generate a final registration status report indicating whether the registration is successful.

## Question 1: Resume Information Extraction System

Regular expressions are used to extract the candidate's name, email, mobile number, technical skills and years of experience from resume text, and to check shortlisting eligibility (minimum 2 years and Python skill).

### Program

```
import re

# Sample resume in text format
resume = """
Name: Rahul Sharma
Email: rahul.sharma@gmail.com
Mobile: 9876543210
Skills: Python, Java, SQL, Machine Learning, NLP
Experience: 3 years
"""

print("RESUME INFORMATION EXTRACTION")
print("-" * 35)

# Extract candidate name
name_match = re.search(r'Name:\s*([A-Za-z ]+)', resume)
name = name_match.group(1).strip() if name_match else "Not Found"

# Identify email address
email_match = re.search(r'[\w\.-]+@[\w\.-]+\.\w+', resume)
email = email_match.group() if email_match else "Not Found"

# Extract 10-digit mobile number
mobile_match = re.search(r'\b[6-9]\d{9}\b', resume)
mobile = mobile_match.group() if mobile_match else "Not Found"

# Detect technical skills
skill_list = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
skills = [
    skill for skill in skill_list
    if re.search(r'\b' + re.escape(skill) + r'\b',
                 resume, re.IGNORECASE)
]

# Extract years of experience
experience = re.search(
    r'Experience:\s*(\d+)\s*years?',
    resume, re.IGNORECASE
)

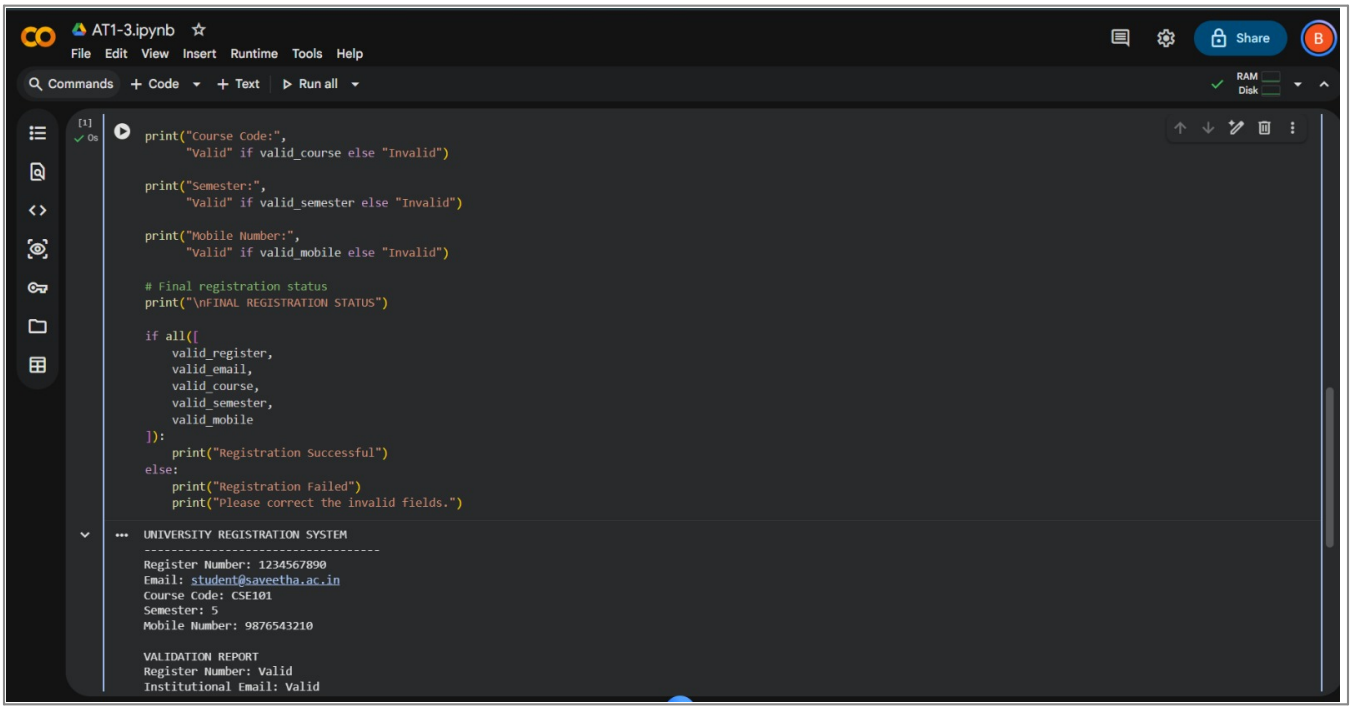
years = int(experience.group(1)) if experience else 0

# Display candidate profile
print("Candidate Name:", name)
print("Email Address:", email)
print("Mobile Number:", mobile)
print("Technical Skills:", ", ".join(skills) if skills else "None")
print("Experience:", years, "years")

# Eligibility check
print("\nELIGIBILITY REPORT")

if years >= 2 and "Python" in skills:
    print("Eligible for Shortlisting")
else:
    print("Not Eligible for Shortlisting")
```

## Output




**Justification:** The program extracts Rahul Sharma, rahul.sharma@gmail.com, 9876543210, the skills Python, Java, SQL, Machine Learning and NLP, and 3 years of experience. Since experience (3) is at least 2 years and Python is present, the candidate is eligible for shortlisting.

## Question 2: Product Search System

A menu-driven product search system using regular expressions supports exact, prefix, suffix, partial and case-insensitive keyword searches, displays all matching products and reports the total number of matches.

### Program

```
import re

# Product database
products = [
    "Apple iPhone 15",
    "Apple MacBook Air",
    "Samsung Galaxy S23",
    "Dell Inspiron Laptop",
    "HP Pavilion Laptop",
    "Lenovo ThinkPad Laptop",
    "Sony Bluetooth Headphones",
    "Boat Wireless Earphones",
    "Logitech Wireless Mouse"
]

print("PRODUCT SEARCH SYSTEM")
print("-" * 30)

while True:
    print("\n1. Exact Keyword Search")
    print("2. Prefix Search")
    print("3. Suffix Search")
    print("4. Partial Keyword Search")
    print("5. Case-Insensitive Search")
    print("6. Display All Products")
    print("7. Exit")

    choice = input("Enter your choice: ").strip()
    matches = []

    if choice == "1":
        keyword = input("Enter exact keyword: ").strip()
        pattern = r'\b' + re.escape(keyword) + r'\b'
        matches = [p for p in products if re.search(pattern, p)]

    elif choice == "2":
        prefix = input("Enter prefix: ").strip()
        pattern = r'^' + re.escape(prefix)
        matches = [p for p in products
                   if re.search(pattern, p, re.IGNORECASE)]

    elif choice == "3":
        suffix = input("Enter suffix: ").strip()
        pattern = re.escape(suffix) + r'$'
        matches = [p for p in products
                   if re.search(pattern, p, re.IGNORECASE)]

    elif choice == "4":
        partial = input("Enter partial keyword: ").strip()
        pattern = re.escape(partial)
        matches = [p for p in products if re.search(pattern, p)]

    elif choice == "5":
        keyword = input("Enter keyword: ").strip()
        pattern = re.escape(keyword)
        matches = [p for p in products
                   if re.search(pattern, p, re.IGNORECASE)]

    elif choice == "6":
        matches = products

    elif choice == "7":
```

```
        print("Exiting Product Search System...")
        break

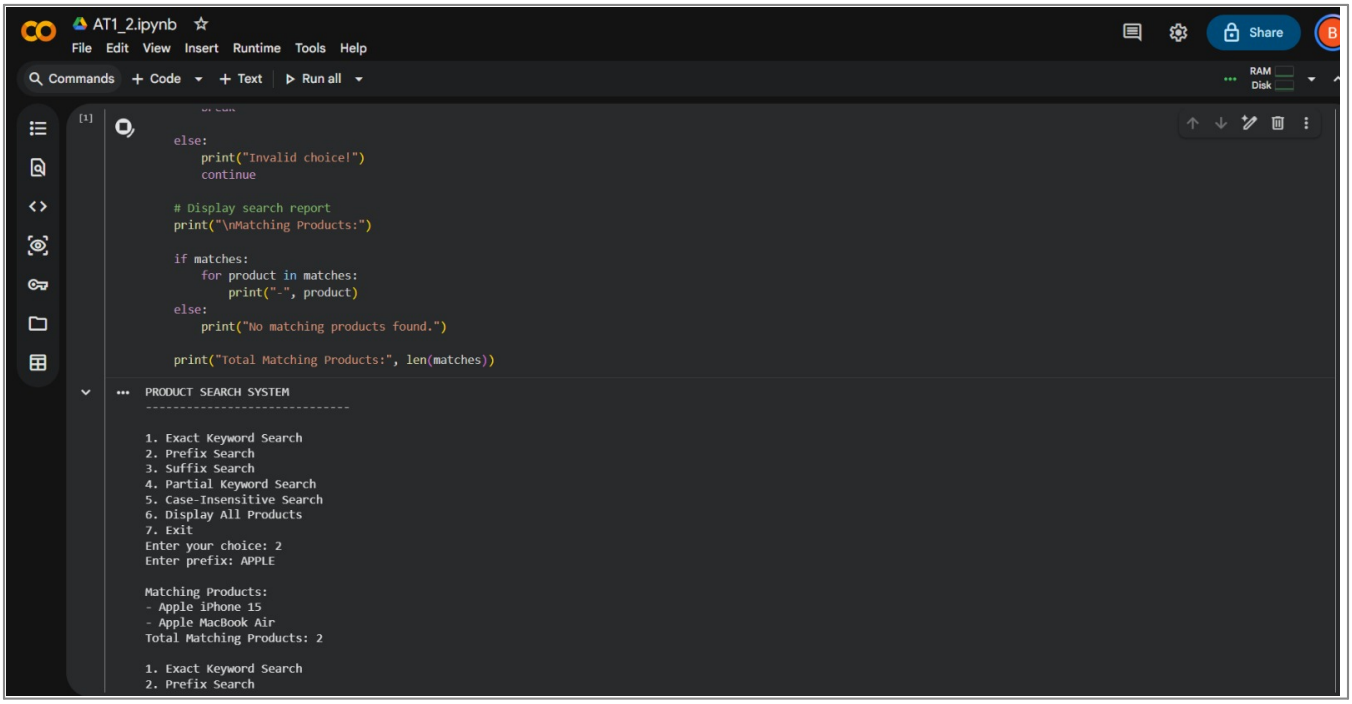
    else:
        print("Invalid choice!")
        continue

    # Display search report
    print("\nMatching Products:")

    if matches:
        for product in matches:
            print("-", product)
    else:
        print("No matching products found.")

    print("Total Matching Products:", len(matches))
```

## Output




**Justification:** Choice 2 (prefix search) with the input APPLE uses the pattern ^APPLE with re.IGNORECASE, so it matches Apple iPhone 15 and Apple MacBook Air. The report lists both products with a total count of 2, and the menu is shown again until Exit is chosen.

## Question 3: University Registration System

Regular expressions validate the register number, institutional email, course code, semester and mobile number. A validation message is displayed for each field and a final registration status is generated.

### Program

```
import re

# Student details to be validated
register_number = "1234567890"
email = "student@saveetha.ac.in"
course_code = "CSE101"
semester = "5"
mobile = "9876543210"

print("UNIVERSITY REGISTRATION SYSTEM")
print("-" * 35)

print("Register Number:", register_number)
print("Email:", email)
print("Course Code:", course_code)
print("Semester:", semester)
print("Mobile Number:", mobile)

# Regex validation rules
valid_register = re.fullmatch(r'\d{10}', register_number) is not None
valid_email = re.fullmatch(r'[\w\.-]+@saveetha\.ac\.in', email) is not None
valid_course = re.fullmatch(r'[A-Z]{3}\d{3}', course_code) is not None
valid_semester = re.fullmatch(r'[1-8]', semester) is not None
valid_mobile = re.fullmatch(r'[6-9]\d{9}', mobile) is not None

# Display validation messages
print("\nVALIDATION REPORT")

print("Register Number:",
      "Valid" if valid_register else "Invalid")

print("Institutional Email:",
      "Valid" if valid_email else "Invalid")

print("Course Code:",
      "Valid" if valid_course else "Invalid")

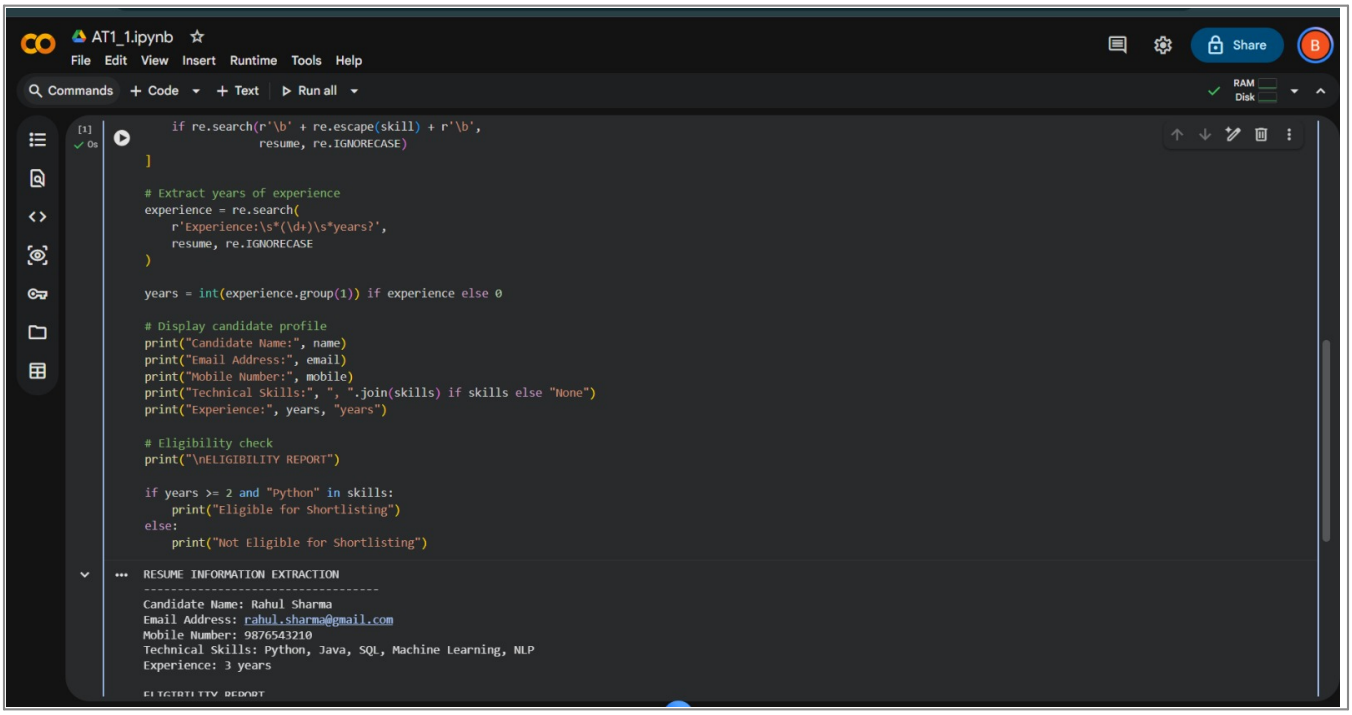
print("Semester:",
      "Valid" if valid_semester else "Invalid")

print("Mobile Number:",
      "Valid" if valid_mobile else "Invalid")

# Final registration status
print("\nFINAL REGISTRATION STATUS")

if all([
    valid_register,
    valid_email,
    valid_course,
    valid_semester,
    valid_mobile
]):
    print("Registration Successful")
else:
    print("Registration Failed")
    print("Please correct the invalid fields.")
```

## Output




**Justification:** Each field matches its pattern: 10-digit register number, email ending with @saveetha.ac.in, course code of 3 capital letters followed by 3 digits (CSE101), semester 1-8, and a 10-digit mobile number starting with 6-9. All fields are valid, so the registration is successful.

## Rubrics for Calculation

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| **Understanding of Problem Context** | 20 | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| **Application of Concepts** | 30 | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy | Fails to apply appropriate concepts |
| **Problem-Solving Approach** | 20 | Logical, well-structured, and efficient approach | Mostly logical approach with minor gaps | Basic approach with some inconsistencies | Illogical or unclear approach |
| **Accuracy of Solution** | 20 | Completely correct solution with proper justification | Mostly correct with minor errors | Partially correct solution | Incorrect or incomplete solution |
| **Clarity** | 10 | Clear, well-organized, and properly explained solution | Understandable with minor clarity issues | Limited clarity and explanation | Poorly presented and difficult to understand |

| Q. No. / Criteria | Understanding of Problem Context | Application of Concepts | Problem-Solving Approach | Accuracy of Solution | Clarity | Sub. Total | Total % |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| | | |
| Signature: _____ | Signature: _____ | Signature: _____ |
| Date: _____ | Date: _____ | Date: _____ |